Discrete Diffusion Tutorial

Forward Noising, Reverse Kernels, and Learning

Roadmap

1. Forward Process
2. Backward Kernels
3. Numerics
4. Learning Reverse Kernels
5. Takeaways

State Space and Histograms

We work on a finite state space $\mathcal{X} = \{1, \ldots, n\}$.
A histogram at time t is

$$h^t \in \mathbb{R}_+^n, \qquad \sum_{i=1}^{n} h_i^t = 1.$$

A forward kernel is row-stochastic:

$$P \in \mathbb{R}_+^{n \times n}, \qquad P\mathbf{1} = \mathbf{1}, \qquad P_{ij} = \mathbb{P}(X_{t+1} = j \mid X_t = i).$$

Forward Dynamics

Definition (Forward recursion)

$$h^{t+1} = P^\top h^t, \qquad t = 0, \ldots, T-1.$$

Proposition 1

If $h^0 \in \Delta_n$ and $P\mathbf{1} = \mathbf{1}$, then each $h^t \in \Delta_n$.

Proof.

Non-negativity is immediate since $P \geq 0$. Also

$$\mathbf{1}^\top h^{t+1} = \mathbf{1}^\top P^\top h^t = (P\mathbf{1})^\top h^t = \mathbf{1}^\top h^t = 1.$$

□

Examples of Forward Kernels

- Local 3-neighbor (random walk style):

$$P_{ij} = \tfrac{1}{3}\mathbf{1}_{j\in\{i-1,i,i+1\}}$$

  (with boundary convention).

- Gaussian convolution:

$$P_{ij} \propto \exp\left(-\frac{(i-j)^2}{2\sigma^2}\right), \qquad \sum_j P_{ij} = 1.$$

- Masking/absorbing (toward a special state n):

$$P_{ii} = 1-\varepsilon, \quad P_{in} = \varepsilon, \quad P_{nn} = 1.$$

Admissible Reverse Kernels

Fix two histograms $a, b \in \Delta_n$.

Definition

$$\mathcal{T}(a, b) = \{K \geq 0 : \ K\mathbf{1} = \mathbf{1}, \ K^\top a = b\}.$$

In diffusion, reverse kernels satisfy

$$Q^t \in \mathcal{T}(h^{t+1}, h^t), \qquad t = 0, \ldots, T-1.$$

So there are exactly T reverse kernels connecting $t+1 \to t$.

Backward Particle Process

Definition (Backward chain)

$$Y_T \sim h^T, \qquad Y_t \sim Q^t(Y_{t+1}, \cdot), \ t = T-1, \dots, 0.$$

Proposition 2

If each $Q^t \in \mathcal{T}(h^{t+1}, h^t)$, then

$$\mathrm{Law}(Y_t) = h^t \quad \textit{for all } t.$$

Proof.

Backward induction: $\mathrm{Law}(Y_T) = h^T$ by initialization. If $\mathrm{Law}(Y_{t+1}) = h^{t+1}$ then

$$\mathrm{Law}(Y_t) = (Q^t)^\top h^{t+1} = h^t.$$

□

Bayes Inverse Kernel

Given forward $P \in \mathcal{T}(h^t, h^{t+1})$, define

$$Q^{\mathrm{B},t}_{ij} = \mathbb{P}(X_t = j \mid X_{t+1} = i).$$

Proposition 3 (Bayes formula + matrix form)

For $h^{t+1}_i > 0$,

$$Q^{\mathrm{B},t}_{ij} = \frac{h^t_j P_{ji}}{h^{t+1}_i}, \qquad Q^{\mathrm{B},t} = \mathrm{diag}(h^{t+1})^+ P^\top \,\mathrm{diag}(h^t).$$

Moreover $Q^{\mathrm{B},t} \in \mathcal{T}(h^{t+1}, h^t)$.

Couplings and Kernels

Define coupling polytope:

$$\Pi(a, b) = \{\pi \geq 0 : \ \pi\mathbf{1} = a, \ \pi^\top \mathbf{1} = b\}.$$

Proposition 4 (Bijection)

$$\Phi_{a,b} : \mathcal{T}(a, b) \to \Pi(a, b), \qquad \Phi_{a,b}(K) = \mathrm{diag}(a)K$$

is bijective on $\mathrm{supp}(a)$ *with inverse*

$$\Phi_{a,b}^{-1}(\pi) = \mathrm{diag}(a)^+\pi.$$

Interpretation: learning Q^t is equivalent to learning a coupling π^t with fixed marginals.

Numerical Setup

We use the same setup as the notebook:

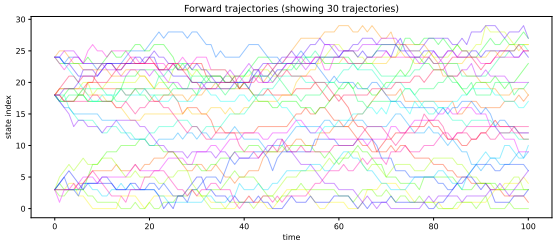
- $n = 30,\ T = 100,\ N = 2500,\ \sigma = 0.75.$
- Initial histogram: 3 spikes at $[0.1, 0.6, 0.8] \cdot n$ with amplitudes $[1, 1.4, 0.6]$ (normalized).
- Forward kernel: Gaussian convolution

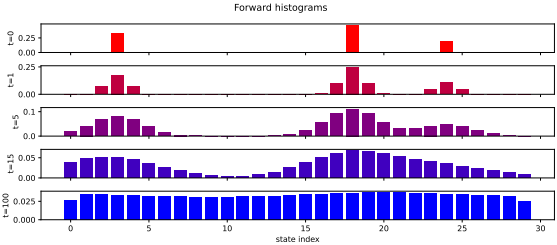
$$P_{ij} \propto \exp\left(-\frac{(i-j)^2}{2\sigma^2}\right).$$

- Reverse kernels: Bayes inversion

$$Q^t = \operatorname{diag}(h^{t+1})^+ P^\top \operatorname{diag}(h^t).$$

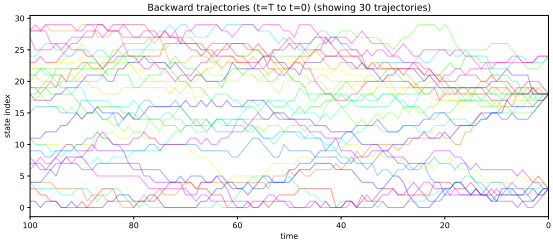
Forward Simulation Results

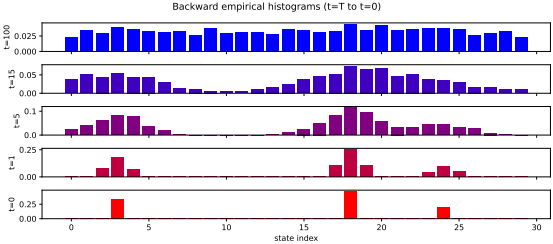




Left: 30 trajectories. Right: snapshots at $t = 0, 1, 5, 15, T$.

Backward Simulation Results





Time is displayed from T (left) to 0 (right) in trajectories.

Learning Objective

From forward samples $(X_{t-1}^{(m)}, X_t^{(m)}, t_m)$, minimize

$$\mathcal{L}(\theta) = -\frac{1}{M} \sum_{m=1}^{M} \log Q_\theta^{t_m - 1}(X_{t_m}^{(m)}, X_{t_m - 1}^{(m)}).$$

- This is cross-entropy on previous-state prediction.
- Population form:

$$\mathcal{R}(\theta) = C_\star + \mathbb{E}\big[\mathrm{KL}(Q_\star^t(i,\cdot) \| Q_\theta^t(i,\cdot))\big].$$

- Optional consistency regularization:

$$\Delta(\theta) = \frac{1}{T} \sum_t \|(Q_\theta^t)^\top h^{t+1} - h^t\|_1.$$

Model Parameterization (Pedagogical)

For 1D geometric histograms, encode

$$x = \frac{i}{n-1} \in [0,1], \qquad \tau = \frac{t}{T} \in [0,1].$$

Input is 2D: (x, τ) (not one-hot in geometric setting).

2-layer MLP

$$(x, \tau) \xrightarrow{\text{Linear } (2 \to p)} \mathrm{ReLU} \xrightarrow{\text{Linear } (p \to n)} \mathrm{Softmax}$$

with $p = 4n$.

If no spatial prior is available, replace x by $\mathrm{onehot}(i)$.

Minimal Python Example

```python
class ReverseNet(nn.Module):
    def __init__(self, n, p):
        super().__init__()
        self.fc1 = nn.Linear(2, p)
        self.fc2 = nn.Linear(p, n)

    def forward(self, x_norm, t_norm):
        z = torch.cat([x_norm, t_norm], dim=1)
        z = F.relu(self.fc1(z))
        return torch.softmax(self.fc2(z), dim=1)

# x_norm = i/(n-1), t_norm = t/T
# loss = -log Q_theta^{t-1}(x_t, x_{t-1}) averaged over samples
```

Takeaways and Practical Checklist

- Forward recursion: $h^{t+1} = P^\top h^t$.
- Reverse admissibility: $Q^t \in \mathcal{T}(h^{t+1}, h^t)$.
- Canonical inverse: Bayes kernel $Q^{\mathrm{B},t} = \mathrm{diag}(h^{t+1})^+ P^\top \mathrm{diag}(h^t)$.
- Learning: optimize NLL on transitions; monitor marginal consistency.
- Numerics: compare true vs learned $\log Q^t_{\lfloor n/2 \rfloor,:}$ at multiple times.